

Gamma Distribution

A continuous random variable X is said to follow a General gamma distribution with parameters $\lambda > 0$ and $k > 0$ if its probability density function is given by

$$f(x) = \begin{cases} \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

We check the above f defines a probability density function.

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_0^{\infty} f(x) dx = \int_0^{\infty} \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)} dx \\ &= \frac{\lambda}{\Gamma(k)} \int_0^{\infty} (\lambda x)^{k-1} e^{-\lambda x} dx \end{aligned}$$

$$\text{put } t = \lambda x \text{ then } dt = \lambda dx \Rightarrow dx = \frac{dt}{\lambda}$$

$$\begin{aligned} \therefore \text{The last integration becomes. } & \frac{\lambda}{\Gamma(k)} \int_0^{\infty} t^{k-1} e^{-t} \frac{dt}{\lambda} \\ &= \frac{1}{\Gamma(k)} \int_0^{\infty} t^{k-1} e^{-t} dt \rightarrow \Gamma(k) \end{aligned}$$

$$= 1$$

Remarks

$$(i) \text{ If } k=1 \text{ then } f(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$\therefore f$ is the probability density function of exponential distribution

$$(ii) \text{ If } \lambda=1 \text{ then } f(x) = \begin{cases} \frac{x^{k-1} e^{-x}}{\Gamma(k)} & , x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

In this we say that X follows Gamma distribution

Mean and variance of Gamma distribution

$$\begin{aligned} E(X^r) &= \int_0^{\infty} x^r f(x) dx = \int_0^{\infty} x^r \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)} dx \\ &= \int_0^{\infty} \frac{\lambda^k x^{r+k-1} e^{-\lambda x}}{\Gamma(k)} dx = \int_0^{\infty} \frac{\lambda^{k-1+1+r-r} x^{r+k-1} e^{-\lambda x}}{\Gamma(k)} dx \\ &= \frac{\lambda^{-r+1}}{\Gamma(k)} \int_0^{\infty} \lambda^{r+k-1} x^{r+k-1} e^{-\lambda x} dx \\ &= \frac{\lambda^{1-r}}{\Gamma(k)} \int_0^{\infty} (\lambda x)^{r+k-1} e^{-\lambda x} dx \rightarrow (*) \end{aligned}$$

Take $\lambda x = t$ then $\frac{dt}{\lambda} = dx$ apply in (*)

we have

$$\begin{aligned} &= \frac{\lambda^{1-r}}{\Gamma(k)} \int_0^{\infty} t^{k+r-1} e^{-t} \frac{dt}{\lambda} \\ &= \frac{\lambda^{-r}}{\Gamma(k)} \Gamma(k+r) \end{aligned}$$

$$\therefore E(X^r) = \frac{\Gamma(k+r)}{\lambda^r \Gamma(k)}$$

$$\text{Mean} = E(X) = \frac{1}{\lambda} \frac{\Gamma(k+1)}{\Gamma(k)} = \frac{1}{\lambda} \frac{k \Gamma(k)}{\Gamma(k)}$$

$$\therefore \text{Mean} = E(X) = \frac{k}{\lambda}$$

$$E(X^2) = \frac{1}{\lambda^2} \frac{\Gamma(k+2)}{\Gamma(k)} = \frac{1}{\lambda^2} \frac{(k+1) k \Gamma(k)}{\Gamma(k)} = \frac{k(k+1)}{\lambda^2}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{k^2}{\lambda^2} + \frac{k}{\lambda^2} - \frac{k^2}{\lambda^2} = \frac{k}{\lambda^2}$$

$$\text{Var}(X) = \frac{k}{\lambda^2}$$

Pb:- In a certain city, the daily consumption of electric power in millions of kilowatt-hours can be treated as a random variable having a general gamma distribution with parameters $\lambda = 1/2$ and $k = 3$.

If the power plant of this city has a daily capacity of 12 millions kilowatt-hours, what is the probability that this power supply will be inadequate on any given day.

Soln:- Let X represent the daily consumption of electric power.

Then the density function of X is

$$f(x) = \frac{\left(\frac{1}{2}\right)^3}{\Gamma(3)} x^2 e^{-x/2} \quad x \geq 0$$

$P(\text{the power supply is inadequate})$

$$= P(X > 12) = \int_{12}^{\infty} f(x) dx$$

$$= \int_{12}^{\infty} \frac{1}{8\Gamma(3)} x^2 e^{-x/2} dx = \frac{1}{16} \int_{12}^{\infty} x^2 e^{-x/2} dx$$

$$= \frac{1}{16} \left[\frac{x^2 e^{-x/2}}{-1/2} - \frac{2x e^{-x/2}}{1/4} + \frac{2 e^{-x/2}}{-1/8} \right]_{12}^{\infty}$$

$$= \frac{1}{16} \left[e^{-6} (288 + 96 + 16) \right] = 0.0625$$